It's not uncommon to see time data incoming from a browser request ultimately recorded in a database as an adjacent day. JavaScript's native `Date` class has its own quirks and pitfalls, which we won't delve into here, but suffice it to say that you should pay close attention to incoming date strings. The JS `Date` API does not support the setting of timezones and will simply use the client's system settings. A user might set a date in a form control as "today," which will be represented with a time of 00:00 hours in the browser's timezone (JS `Date` objects use Unix timestamps as their internal representation of time).

A sensible approach for handling time data in web apps is to use Unix timestamps in as many places as possible and treat the rendering of dates and times as a localization problem, to be solved only once you know as much as possible about the user's location and preferences. Most of the technology related to PHP will offer some API or function call to obtain a timestamp. JavaScript `Date` objects are implicit timestamps, and MySQL offers the `TIMESTAMP` data type and a `UNIX_TIMESTAMP()` function, which is equivalent to PHP's `time()`. MySQL also offers `DATE` and `DATETIME` types, which are simply formatted strings much like those produced by PHP's `DateTime::format()` method, but with the important difference that there is *no* timezone support for these types. Tread carefully around storing time with MySQL and its brethren (such as MariaDB).

The handling of daylight saving time (DST) presents challenges when performing arithmetic, and so too do leap years. Take a look at the following:

```
date_default_timezone_set('Europe/London');

$winterIsComing = new DateTime('26-OCT-2025');
print $winterIsComing->format(DateTime::COOKIE) . PHP_EOL;

$winterIsComing->add(new DateInterval('PT3H'));
print $winterIsComing->format(DateTime::COOKIE);

Output:
Sunday, 26-Oct-2025 00:00:00 BST
Sunday, 26-Oct-2025 02:00:00 GMT
```

The interval format `PT3H` specifies an addition of three hours, but the difference in time appears to be only two hours if we just look at the date and time. Of course, we've switched timezone due to DST – British Summer Time to GMT – thanks to it being the last Sunday in October (when the UK "falls back"). Daylight savings changes like these are handled pretty well by PHP; here's an inversion of the example above…